

SOFTWARE DESIGN DOCUMENT (SDD)

Project Title

Sudanile Kitchen Mobile Application

Prepared By

WATUMANDE ELIWA WATULEVI SAMUEL

ID: SWATEL2221

Major: Software Engineering

Supervisor

Mr. Felix Chepsiror

Version

1.0

Date

March 2026

1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture, design, database structure, modules, interfaces, and implementation details of the Sudanile Kitchen Mobile Application.

The document serves as a blueprint for developers, testers, project supervisors, and future maintainers by providing detailed technical information about the system.

1.2 Project Overview

Sudanile Kitchen is a mobile application designed to preserve, promote, and share authentic South Sudanese recipes. The system provides users with access to traditional dishes, cooking instructions, ingredient information, user reviews, and personalized recipe collections.

The application addresses the challenge of preserving South Sudanese culinary knowledge by digitizing recipes and making them accessible worldwide.

1.3 Scope

The system will provide:

- User registration and authentication
- Recipe browsing and searching
- Recipe details with cooking instructions
- User reviews and ratings
- Favorite recipe management
- Recipe submission functionality
- User profile management

Out of Scope

The following features will NOT be implemented in this version:

- Social media integration
 - Online food ordering
 - Recipe video streaming
 - AI recipe recommendation engine
 - Future enhancements listed in the proposal
-

2. SYSTEM ARCHITECTURE

2.1 Architecture Style

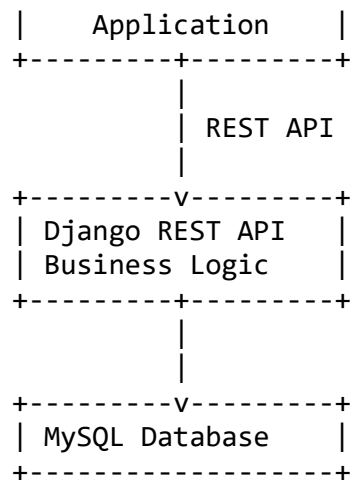
The system follows a **Client-Server Architecture**.

Components

1. Flutter Mobile Application (Frontend)
 2. Django REST API (Backend)
 3. MySQL Database
-

2.2 High-Level Architecture

```
+-----+
|  Flutter Mobile  |
```



3. TECHNOLOGY STACK

Frontend

- Flutter
- Dart

Backend

- Python
- Django
- Django REST Framework

Database

- MySQL

Development Tools

- Visual Studio Code
 - Postman
 - Git
 - GitHub
-

4. PROJECT STRUCTURE

Backend Structure

backend/

```
|— config/
|   |— settings.py
|   |— urls.py
|   |— wsgi.py
|
|— users/
|— recipes/
|— reviews/
|— favorites/
|— submissions/
|
|— requirements.txt
|— manage.py
```

Mobile Structure

mobile/

```
|— lib/
|
|   |— models/
|   |   |— recipe.dart
|   |   |— user.dart
|   |   |— review.dart
|   |
|   |— services/
|   |   |— api_service.dart
|   |   |— auth_service.dart
|   |   |— recipe_service.dart
|   |
|   |— screens/
|   |   |— splash_screen.dart
|   |   |— login_screen.dart
|   |   |— register_screen.dart
|   |   |— home_screen.dart
|   |   |— recipe_detail_screen.dart
|   |   |— search_screen.dart
|   |   |— favorites_screen.dart
|   |   |— profile_screen.dart
|   |   |— submit_recipe_screen.dart
```

```
| | | widgets/  
| | | | recipe_card.dart  
| | | | category_card.dart  
| | | | search_bar.dart  
  
| | | main.dart
```

5. MODULE DESIGN

5.1 User Management Module

Purpose

Handles registration, authentication, and profile management.

Functions

- Register User
- Login User
- Logout User
- Update Profile
- View Profile

Backend Components

```
users/  
| models.py  
| serializers.py  
| views.py  
| urls.py
```

Inputs

- Username
- Email
- Password

Outputs

- User Account
 - Authentication Token
-

5.2 Recipe Management Module

Purpose

Stores and manages recipes.

Functions

- Add Recipe
- Edit Recipe
- Delete Recipe
- View Recipe
- Browse Recipes

Backend Components

```
recipes/  
├── models.py  
├── serializers.py  
├── views.py  
└── urls.py
```

5.3 Search Module

Purpose

Allows users to search for recipes.

Functions

- Search by Recipe Name
- Search by Category
- Search by Ingredient

Input

Search keyword

Output

Matching recipes

5.4 Reviews Module

Purpose

Allows users to rate and review recipes.

Functions

- Add Review
- Edit Review
- Delete Review
- View Reviews

Backend Components

```
reviews/  
├── models.py  
├── serializers.py  
├── views.py  
└── urls.py
```

5.5 Favorites Module

Purpose

Allows users to save preferred recipes.

Functions

- Add Favorite
- Remove Favorite
- View Favorites

Backend Components

```
favorites/  
├── models.py  
├── serializers.py  
├── views.py  
└── urls.py
```

5.6 Recipe Submission Module

Purpose

Allows users to contribute recipes.

Functions

- Submit Recipe
- View Submission Status

Backend Components

```
submissions/  
├── models.py  
├── serializers.py  
├── views.py  
└── urls.py
```

6. DATABASE DESIGN

6.1 Entities

Users

Attribute	Type
user_id	PK
username	VARCHAR
email	VARCHAR
password	VARCHAR
profile_image	VARCHAR
date_joined	DATETIME

Recipes

Attribute	Type
recipe_id	PK
title	VARCHAR
description	TEXT
ingredients	TEXT
instructions	TEXT
category	VARCHAR
image_url	VARCHAR
created_by	FK
created_at	DATETIME

Reviews

Attribute	Type
review_id	PK
recipe_id	FK
user_id	FK
rating	INTEGER
comment	TEXT
created_at	DATETIME

Favorites

Attribute	Type
favorite_id	PK
user_id	FK
recipe_id	FK
created_at	DATETIME

Submissions

Attribute	Type
submission_id	PK
user_id	FK
title	VARCHAR
ingredients	TEXT
instructions	TEXT
status	VARCHAR
submitted_at	DATETIME

6.2 Entity Relationships

User → Recipe

One user can create many recipes.

User (1) ----- (M) Recipe

User → Review

One user can create many reviews.

User (1) ----- (M) Review

Recipe → Review

One recipe can have many reviews.

Recipe (1) ----- (M) Review

User → Favorite

One user can have many favorites.

User (1) ----- (M) Favorite

Recipe → Favorite

One recipe can appear in many favorites.

Recipe (1) ----- (M) Favorite

User → Submission

One user can submit many recipes.

User (1) ----- (M) Submission

7. API DESIGN

Authentication Endpoints

Method	Endpoint
POST	/api/users/register
POST	/api/users/login
GET	/api/users/profile

Recipe Endpoints

Method	Endpoint
GET	/api/recipes
GET	/api/recipes/{id}
POST	/api/recipes
PUT	/api/recipes/{id}
DELETE	/api/recipes/{id}

Review Endpoints

Method	Endpoint
--------	----------

POST	/api/reviews
------	--------------

GET	/api/reviews/{recipe_id}
-----	--------------------------

Favorite Endpoints

Method	Endpoint
--------	----------

POST	/api/favorites
------	----------------

DELETE	/api/favorites/{id}
--------	---------------------

GET	/api/favorites
-----	----------------

Submission Endpoints

Method	Endpoint
--------	----------

POST	/api/submissions
------	------------------

GET	/api/submissions
-----	------------------

8. USER INTERFACE DESIGN

Splash Screen

Purpose:

- Display application logo.
 - Load application resources.
-

Login Screen

Features:

- Email input
 - Password input
 - Login button
 - Register link
-

Home Screen

Features:

- Recipe categories
 - Search bar
 - Featured recipes
 - Recipe cards
-

Recipe Detail Screen

Features:

- Recipe image
 - Ingredients list
 - Cooking instructions
 - Reviews section
 - Add to Favorites button
-

Favorites Screen

Features:

- Saved recipes
 - Remove favorite option
-

Profile Screen

Features:

- User information
 - Edit profile
 - View submitted recipes
-

Submit Recipe Screen

Features:

- Recipe title
- Ingredients

- Instructions
 - Category
 - Submit button
-

9. SECURITY DESIGN

The system will implement:

Authentication

- JWT Authentication

Password Security

- Password Hashing

API Security

- Token-Based Authorization
- Input Validation

Database Security

- Foreign Key Constraints
 - SQL Injection Protection through Django ORM
-

10. TESTING STRATEGY

Unit Testing

Testing individual modules:

- User Module
 - Recipe Module
 - Review Module
 - Favorites Module
 - Submission Module
-

Integration Testing

Testing interaction between:

- Flutter Application

- Django REST API
 - MySQL Database
-

System Testing

Testing the complete application workflow.

User Acceptance Testing

Testing by end users to ensure requirements are satisfied.

11. DEPLOYMENT DESIGN

Backend Deployment

- Django REST API
- Hosted on cloud server

Database Deployment

- MySQL Server

Mobile Deployment

- Android APK generated from Flutter
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12. CONCLUSION

The Sudanile Kitchen Mobile Application is designed to digitally preserve and promote South Sudanese culinary heritage through a modern mobile platform. The system architecture, database design, APIs, and user interface components provide a scalable, secure, and maintainable solution for recipe discovery, management, and cultural preservation.